NASA SPACE APPS CHALLENGE 2026

Harmonization of MODIS and VIIRS Hot Spots

Deep Research, Public Data Catalog, Scientific Method, UI Plan, Open-Source References, and Hackathon Execution Guide

CORE DATA	CORE OUTPUT	BUILD WINDOW
NASA FIRMS MODIS + VIIRS	**Harmonized Burning Calendar**	**Research pack 3 Oct 2026**

Deliverable scope A comprehensive practical catalog of publicly accessible resources directly useful for this challenge. It is intentionally broader than the minimum MVP, but separates must-use core data from optional enrichment so the team does not lose time during the hackathon.

Prepared for team sharing. All resource names in the report are clickable hyperlinks. Public availability, product versions, and service status can change; re-check source pages when downloading.

Research snapshot: 3 October 2026

Executive summary

The challenge is a sensor-harmonization problem before it is a mapping problem. MODIS provides a long record at roughly 1 km nominal fire-pixel resolution; VIIRS provides a newer record at roughly 375 m and detects many smaller hotspots. Raw hotspot counts therefore create a false discontinuity when the observing system changes. A strong solution should explicitly correct, quantify, and visualize that discontinuity.

Recommended product Build a global web application whose flagship output is a 2000-to-present Burning Activity Calendar for a selected area of interest (AOI). Pair the calendar with a Sensor Lab that shows raw MODIS vs VIIRS, the overlap-based calibration, uncertainty, and validation. Use Bangladesh as the first demonstration AOI and at least one contrasting international AOI.

- **Must-use data:** NASA FIRMS standard/archive MODIS Terra/Aqua and VIIRS S-NPP/NOAA-20/NOAA-21 active-fire records.
- **Strong validation/context:** MCD64A1 and VNP64A1 burned area; land-cover stratification; optional GlobFire/Global Fire Atlas event products.
- **Best scientific differentiator:** resolution-aware, season-aware and land-cover-aware overlap calibration instead of a single global count multiplier.
- **Best UX differentiator:** a calendar first, then map; every result exposes sensor mix, uncertainty, source/version, and why a month is considered unusual.
- **What not to claim:** do not call every hotspot a wildfire and do not present a historical anomaly score as an exact future-fire prediction.

Hackathon timing from the provided Bangladesh submission portal screenshot: 240-second video and a publicly accessible source-code link, with the displayed submission end of 07 Oct 2026, 11:59 PM. Re-check the portal timezone and status before final submission.

Fast decision: what to build

Layer	MVP	Stretch if time remains
Data	Precomputed historical standard FIRMS + live/NRT endpoint for recent days	Additional weather, land cover, population exposure, burned-area event validation
Science	Monthly/seasonal cross-sensor calibration in overlap period; holdout validation	Cell-level robust regression/GAM or calibrated gradient boosting with uncertainty
UI	AOI selector, map, 26-year calendar, raw-vs-harmonized sensor panel, anomaly view	Fire-event clustering/replay, export/report mode, contextual layers
Demo	Bangladesh + one international contrast region	Global free-draw AOI with cached on-demand aggregation
AI/ML	Only where it solves calibration/anomaly or classification	Narrative assistant only if grounded in computed metrics and source metadata

Report map

1. Challenge interpretation
2. Core NASA/FIRMS data you should use
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How to use this pack If the team has only 30 minutes, read Sections 1, 2, 5, 7 and 12. If you own the data/science work, also read Sections 3, 4, 6 and 11. If you own the frontend/demo, focus on Sections 7, 8, 9 and 13.

1. Challenge interpretation

NASA describes the task as building a web application that harmonizes active-fire hotspot records from different sensors into a burning-activity calendar, so early-warning responders, scientists and land managers can examine historical patterns, unusual conditions and critical periods inside a selected AOI.

Official 2026 challenge page - Open link - Primary challenge reference

What "harmonize" means in practice

- **Do not concatenate rows and stop.** VIIRS sees much smaller/higher-resolution thermal anomalies than MODIS; raw counts are not a stable long-term measure.
- **Create a common analytical unit.** Aggregate both sensors to the same AOI/grid and time unit before calibration.
- **Use the overlap period.** From 2012 onward MODIS and S-NPP VIIRS overlap, allowing empirical calibration rather than guessed conversion factors.
- **Preserve sensor provenance.** The user should always be able to see whether a historical value was observed directly, calibrated from MODIS, or generated from a mixed-sensor period.
- **Expose uncertainty.** A harmonized historical estimate is not identical to a direct modern VIIRS observation. Show confidence/uncertainty rather than hiding this difference.

Judge-facing one-liner "We turn two incompatible satellite eras into one transparent, uncertainty-aware fire-activity calendar - without pretending a 1 km MODIS detection is the same thing as a 375 m VIIRS detection."

2. Core NASA/FIRMS data you should use

These are the primary datasets. The MVP can be built entirely from the first five rows; the rest improve interpretation and validation.

Dataset	Resolution	Coverage	Access	Use in project	Link
MODIS Terra C6.1 active fire	1 km nominal	01 Nov 2000-present	CSV/JSON/SHP; standard + NRT	Historical backbone; morning overpass	FIRMS archive
MODIS Aqua C6.1 active fire	1 km nominal	03 Jul 2002-present	CSV/JSON/SHP; standard + NRT	Adds afternoon overpass; combine or model separately	MODIS fields
VIIRS S-NPP active fire	375 m nominal	20 Jan 2012-present*	CSV/JSON/SHP; standard + NRT	Primary overlap sensor; stronger small-fire response	VIIRS fields
VIIRS NOAA-20 active fire	375 m nominal	Archive lists 01 Apr 2018-present	CSV/JSON/SHP; standard + NRT	Modern redundancy / denser observations	FIRMS archive
VIIRS NOAA-21 active fire	375 m nominal	17 Jan 2024-present	CSV/JSON/SHP; NRT and archive availability	Newest VIIRS stream; future continuity	FIRMS active fire
MCD64A1 V6.1 burned area	500 m monthly	Nov 2000-present	Raster / Earth Engine	Secondary validation of season/timing and burned extent	Dataset
VNP64A1 V2 burned area	500 m monthly	Jan 2012-present	Raster / Earth Engine	VIIRS-era burned-area comparison/validation	Dataset
NASA GIBS fire layers	Visualization tiles	Historical layers by product	WMTS/WMS/MVT	Fast map layers/fallback visualization; not a substitute for tabular science data	FIRMS GIBS guide

*Operational note: FIRMS states that delivery of S-NPP satellite data products will cease on 1 Nov 2026 and recommends transition to NOAA-20 and NOAA-21. For your 2026 hackathon this mainly affects future-proofing: make the source registry configurable and avoid hard-coding S-NPP as the only VIIRS sensor.

Metadata inconsistency to document The FIRMS archive page currently lists NOAA-20 coverage from 01 Apr 2018, while the VIIRS field-description page displays 01 Jan 2020. Use the archive/API availability for operational download windows, store the source and version in your provenance table, and do not silently assume coverage.

FIRMS fields worth keeping

Field	MODIS	VIIRS	Why keep it
latitude / longitude	Center of 1 km fire pixel	Center of nominal 375 m pixel	Spatial aggregation and clustering
acq_date + acq_time	UTC acquisition	UTC acquisition	Daily/overpass matching and deduplication
satellite	Terra / Aqua	S-NPP / NOAA-20 / NOAA-21	Sensor provenance and overlap modeling
confidence	0-100	low / nominal / high	Quality filtering and uncertainty
frp	MW	MW	Intensity proxy; useful alongside counts
scan / track	Actual pixel dimensions vary across scan	Actual pixel dimensions vary across scan	Geometry/view-angle proxy
daynight	D/N	D/N	Diurnal patterns; nighttime sensitivity differences

Field	MODIS	VIIRS	Why keep it
type	Standard product only	Standard product/version dependent	Filter/flag vegetation, volcano/static/offshore thermal sources

Data-quality rule For historical analysis, prefer FIRMS standard (science-quality) data once available. NASA advises that NRT data are later replaced by standard data, typically after a lag of a few months. Use NRT only for the "what is happening now" panel.

3. Supporting datasets and validation layers

Use these selectively. They should explain or validate the fire calendar, not overwhelm the core challenge.

Resource	Coverage / form	Best use	Link
MCD12Q1 V6.1 Land Cover	500 m yearly; 2001-2024	Stratify calibration by cropland/forest/grassland; explain seasonal burn regimes	Open
ESA WorldCover 2021	10 m global land cover	Fine modern land-cover context; useful for Bangladesh AOIs	Open
MOD11A1 V6.1 LST	1 km daily	Temperature context; MODIS-era consistent surface-temperature layer	Open
VNP21A1D V2 LST	1 km daily	VIIRS-era day land-surface-temperature context	Open
NASA EONET v3	Event API	Contextual wildfire-event labels and links; useful for current event triage, not full ground truth	Open
GPM IMERG precipitation	Half-hourly to monthly	Rainfall/dryness context and explanatory overlays	Open
NASA-USDA global soil moisture	0.25 degree	Dryness/soil-moisture context	Open
MERRA-2 reanalysis	Hourly/monthly	Wind, humidity and atmospheric context when a full reanalysis is needed	Open
NASA POWER Daily API	Point-based daily meteorology	Fast weather enrichment for selected locations	Open
GFED5 / GFED5.1	Gridded burned area/emissions	Long-run fire/emissions comparison at coarser scale	Open
Global Fire Atlas	Individual fires, 2003-2016	Ignition, size, duration, spread; event-level benchmark	Open
GlobFire v2	Fire event perimeters 2001-2021	Compare cluster/event segmentation with MCD64A1-derived events	Open
geoBoundaries	Global admin boundaries	Clean Bangladesh ADM0/ADM1/ADM2 and global AOI selection	Open
WorldPop	Population grids	Optional exposure indicator for people near hotspot clusters	Open
ERA5-Land	Land reanalysis	Weather/drought enrichment if the team can handle CDS downloads	Open

Validation caution MCD64A1 and products derived from it (including GlobFire and Global Fire Atlas) are extremely useful, but they are not fully independent truth for active-fire detection. MCD64A1 itself uses MODIS active-fire information. Call them secondary/consistency validation and be precise about that limitation.

4. Data access, APIs and tooling

4.1 FIRMS access paths

Path	What it gives you	Access	Link
Archive Download	Historical standard/science-quality CSV, JSON or shapefile; country/year workflows	Earthdata login/email workflow	Open
Active Fire downloads	Last 24h, 48h, 7d; recent text/SHP/KML	Earthdata login for some downloads	Open
Area API	CSV by sensor, bbox, 1-5 day range and optional date	Free FIRMS MAP_KEY	Open
MAP_KEY	Free key; current page states 5000 transactions / 10 minutes	Email signup	Open
Data Availability API	Check SP/NRT date availability by sensor	MAP_KEY	Open
Fire Data Academy	NASA notebooks/tutorials for ingestion, subsetting, projections, API and visualization	Free	Open
GIBS	High-rate map visualization through WMTS/WMS/MVT; FIRMS page recommends it for large transaction rates	Keyless/public for many services	Open
Worldview	Human visual inspection of satellite imagery and thermal anomaly layers	Free web app	Open

4.2 Earth Engine shortcuts (excellent for prototyping)

Collection	Earth Engine ID	Use	Link
MODIS NRT fire	FIRMS	Fast daily raster prototype; NRT, not preferred for final historical science	Catalog
VIIRS S-NPP NRT	NASA/LANCE/SNPP_VIIRS/C2	375 m NRT raster prototype	Catalog
VIIRS NOAA-20 NRT	NASA/LANCE/NOAA20_VIIRS/C2	375 m NRT raster prototype	Catalog
MCD64A1 V6.1	MODIS/061/MCD64A1	Monthly burned-area validation	Catalog
VNP64A1 V2	NASA/VIIRS/002/VNP64A1	Monthly VIIRS burned-area validation	Catalog

Recommended access strategy Historical: download/precompute standard FIRMS once. Live demo: Area API for recent hotspots. Basemap/visual imagery: GIBS. Validation/context: Earth Engine where it saves time. This prevents the live website from reprocessing 26 years of points.

5. Scientific harmonization method

The method below is designed to be defensible within a hackathon: transparent baseline first, more flexible model only if validation proves it helps.

5.1 Why raw counts fail

VIIRS 375 m can detect substantially more small hotspots than MODIS 1 km, and the difference also depends on land cover, fire intensity, season, view geometry and time of overpass. Published comparisons show that relationships in FRP and detection rates vary by scale and ecosystem. A single universal "VIIRS = k x MODIS" coefficient is therefore easy to implement but scientifically fragile.

5.2 Recommended pipeline

1. Ingest and freeze provenance: Download standard FIRMS historical records. Add source product, collection/version, satellite, acquisition UTC and ingestion date to every row.

2. Normalize schema: Map all sensors into one canonical table. Keep raw fields unchanged and add derived columns separately.

3. AOI and spatial grid: For a fast MVP: aggregate at AOI-day/month. Better: use H3 or a 5 km equal-area-ish working grid so both sensors are compared on the same support.

4. Quality filtering: Flag low confidence and non-vegetation/static thermal sources where product/version permits. Do not silently delete; keep counts before/after filtering.

5. Deduplicate / cluster: Within each sensor/day, cluster nearby detections if the UI presents "fire systems" rather than raw pixels. DBSCAN/HDBSCAN is reasonable, but keep raw pixel counts available.

6. Build overlap pairs: 2012-present is the calibration era. Aggregate MODIS and VIIRS at matched AOI/grid-time units. Where feasible use near-simultaneous overpasses; otherwise use day/month units and mark this limitation.

7. Transparent baseline: Estimate robust seasonal scale factors using medians or a robust regression, preferably stratified by month and land cover. This creates a VIIRS-equivalent historical MODIS activity estimate.

8. Optional improved model: Fit a robust GLM/GAM/Huber/negative-binomial model using MODIS count, MODIS FRP, month, land cover and scan/track/view proxies. Try gradient boosting only if it improves held-out years without destroying interpretability.

9. Build activity metrics: Keep at least two harmonized outputs: detection activity and FRP activity. Convert to monthly percentile/z-score only for intuitive calendar display.

10. Quantify uncertainty: Bootstrap calibration years or report fold-to-fold error/CV. Mark pre-2012 values as model-adjusted, not direct VIIRS observations.

11. Validate: Hold out recent years and compare fire-season timing/events against burned-area products/event data. Report MAE/RMSE/correlation and seasonal peak agreement.

12. Precompute: Write compact Parquet/JSON tables consumed by the web app. Live API calls should cover recent observations, not the whole historical archive.

5.3 Baseline formulas you can explain in 20 seconds

Seasonal calibration baseline For month m and overlap years y, compute a robust ratio r_m = median[(VIIRS activity + epsilon) / (MODIS activity + epsilon)]. For pre-VIIRS years, estimate VIIRS-equivalent activity as MODIS x r_m. Use cross-validation and show the variability of r_m.

A more stable alternative for sparse cells is a regression on log1p counts/FRP rather than raw ratios. For the calendar, display a percentile relative to that AOI and calendar month: e.g., "92nd historical percentile for April". That avoids implying a universal physical unit when the product is really a normalized activity indicator.

5.4 What to call the output

- **Harmonized hotspot activity**Good default term; clearly tied to observation.
- **VIIRS-equivalent historical activity**Good for calibrated pre-2012 estimates if that is genuinely your target scale.

- **Burning Activity Percentile**Excellent UI metric if computed relative to an AOI/month climatology.
- **Avoid: wildfire count**One satellite hotspot is not necessarily one wildfire, and multiple pixels can belong to one fire event.
- **Avoid: fire probability / prediction**Unless you actually train and validate a forecasting model with independent labels and weather predictors.

6. Validation and evaluation

A good hackathon solution should show that harmonization improves continuity across the sensor transition rather than only producing attractive maps.

Test	How to run it	What success looks like
Temporal holdout	Fit calibration on earlier overlap years, test on later years (e.g., hold out 2024-2025)	Lower error than raw MODIS count and stable monthly bias
Sensor continuity	Plot annual/monthly series before and after 2012 transition	Reduced artificial jump while preserving real seasonal variation
Seasonality	Compare monthly climatology before/after calibration	Peak months remain physically plausible and consistent
Land-cover slices	Evaluate forest, cropland, grassland separately	No single class dominates errors; document where it does
FRP consistency	Compare aggregated FRP by sensor on matched units	Strong relationship with residuals reported; do not assume exact equality
Burned-area timing	Compare active-fire peak dates with MCD64A1/VNP64A1 burn dates	Reasonable timing agreement; note products are not independent truth
Event-level check	Compare a few clusters with GlobFire/Global Fire Atlas/EONET/imagery	Known large events appear in expected time/place
Robustness	Run with/without low-confidence or static-source filters	Key calendar conclusions do not hinge on one arbitrary filter

Metrics worth putting in the demo

- MAE or median absolute error on held-out overlap units
- RMSE only if large outliers are meaningful to your model
- R-squared or Spearman correlation for cross-sensor agreement
- Monthly peak agreement / seasonal rank correlation
- Calibration uncertainty (bootstrap interval, CV, or fold spread)
- Coverage fraction: days/cells with valid observations, where feasible

7. Product and UI blueprint

The UI should make the harmonization problem visible, not hide it. A judge should understand the sensor mismatch within the first minute of the demo.

Screen	What the user does	What it shows	Why it matters
1. AOI Explorer	Search Bangladesh/district, choose preset, or draw polygon	Current/recent MODIS + VIIRS hotspots, sensor legend, date slider	Immediate spatial context and interactive entry point
2. Burning Calendar	Choose year range and month/day view	26-year x 12-month heatmap of harmonized activity; click any cell	Directly answers the challenge deliverable
3. Sensor Lab	Toggle Raw / Harmonized	MODIS vs VIIRS overlap scatter, raw counts/FRP, correction/model, uncertainty	Proves you solved harmonization rather than concatenation
4. Anomaly View	Select current month or historical event	Percentile/z-score versus same-month climatology; "unusual" flags	Decision-oriented insight without overclaiming forecast skill

Screen	What the user does	What it shows	Why it matters
5. Context	Toggle land cover/rainfall/LST/population	Why a region/month behaves differently	Adds explainability and local relevance
6. Method & Provenance	Inspect product/version/data date	Source links, sensor era, standard-vs-NRT status, caveats	Scientific trust and reproducibility
7. Export (stretch)	Download chart/CSV/summary	AOI report with source/coverage metadata	Useful to researchers and responders

Suggested Bangladesh demo Make Bangladesh a first-class preset, but keep the engine global. Useful AOI presets could include Chattogram Hill Tracts, Sundarbans region, Sylhet and a north-west agricultural-burn region - then include one contrasting global region (e.g., Mediterranean, boreal or tropical forest) to demonstrate generalization.

UI details judges will notice

- **Sensor transparency:** color/toggle by Terra, Aqua, S-NPP, NOAA-20, NOAA-21.
- **Source badge:** "Standard archive" versus "NRT" should be visible.
- **Calendar legend:** percentile or normalized activity, not raw pixel count across eras.
- **Click-through:** a calendar cell should load the corresponding map/month and show raw sensor values.
- **Uncertainty badge:** High/medium/low or numeric interval with a tooltip explaining why.
- **No-data state:** different from zero activity; cloud/overpass limitations make this distinction important.
- **Accessible colors and labels:** do not depend on red/green alone; label high/low and sensor names.

8. System architecture and data model

A hackathon-friendly architecture keeps heavy geospatial work offline/precomputed and serves compact summaries to the browser.

Recommended stack Python + Polars/Pandas + GeoPandas/H3 for ETL; DuckDB or Postgres/PostGIS for aggregation; FastAPI for a thin API; Next.js/React frontend; MapLibre GL or Leaflet for maps; ECharts/Plotly for calendar and diagnostic charts. GIBS can provide satellite/thermal-anomaly visual layers.

Canonical raw table

Column	Type / example	Notes
sensor	modis / viirs	Logical family
satellite	Terra / Aqua / S-NPP / NOAA20 / NOAA21	Keep exact source
datetime_utc	timestamp	Derived from acq_date + acq_time
lat, lon	float	Original centroid
frp_mw	float	Do not impute blindly
confidence_raw	string/number	Preserve sensor-native representation
confidence_class	low / nominal / high	Normalized derived field
daynight	D/N	
type_raw	integer/null	Version-dependent
scan_km, track_km	float/null	Useful geometry proxy
source_version	e.g., 6.1 / 2.x	Critical provenance

Column	Type / example	Notes
processing_class	standard / nrt / rt / urt	Do not mix silently

Derived calendar table

Column	Meaning
aoi_id / grid_id	Selected AOI or common working grid
date / year / month	Calendar grain
modis_count, viirs_count	Raw sensor-specific detection totals
modis_frp, viirs_frp	Raw sensor-specific FRP sums/robust summaries
cluster_count	Optional distinct event/system approximation
harmonized_count	Calibrated common-scale activity
harmonized_frp	Calibrated FRP-style activity if modeled
activity_percentile	0-100 within AOI/month climatology
anomaly_z	Standardized anomaly (robust if possible)
uncertainty	Bootstrap interval/CV/fold spread
source_mix	Which satellites contributed
quality_flags	No-data, low coverage, NRT, sparse calibration, etc.

9. Open-source and similar projects

Competition ethics A public repository from a current 2026 competitor can be studied as a benchmark, but do not copy its code, wording, UI, datasets, or claimed results into your submission. Build an independently designed solution and cite any code you actually reuse according to its license.

Project	Why inspect it	License note	Link
IGNIS - current 2026 same-challenge public submission	Very close benchmark: 26-year calendar, FIRMS/GIBS, DBSCAN, FRP, anomaly/trend, Bangladesh module, open science assets. Useful to understand the bar; differentiate scientifically and visually.	MIT	Repository
MODIS-VIIRS-fires (Meredith Franklin)	R pipeline reading FIRMS MODIS/VIIRS CSV, converting to UTM and using HDBSCAN to identify persistent combustion sources; directly relevant clustering reference.	MIT	Repository
hotspot-disambiguation (LINKS)	Code/models for classifying satellite hotspots as wildfires using EFFIS cross-referencing; pairs with the HF dataset below.	Check repo	Repository
wildfire dataset pipeline (JasmineW6214)	Reusable 1 km daily multi-source wildfire dataset-generation pipeline using Earth Engine; helpful for grid/feature engineering.	See repo	Repository

Project	Why inspect it	License note	Link
viirs-modis-fire-map	Older MODIS/VIIRS fire map example; useful for timeline/vector-tile visualization patterns.	Check repo	Repository
fireMap	FIRMS + weather/FWI-style map application; useful product reference, not core harmonization science.	MIT reported	Repository
FireSight	Client-side FIRMS NRT mapping/dedup example; useful for lightweight frontend ideas.	Check repo	Repository
Call for Code fires-api-nodejs	Fire report ingestion/API patterns and GeoJSON backend example.	Check repo	Repository
fire-map (kjaymiller)	FastAPI/Postgres-style FIRMS history architecture reference.	Check repo	Repository

Current Space Apps concept signals

Public team pages already show multiple teams converging on ETL, deduplication, GIS and historical calendars. Celestial X, for example, publicly describes an automated FIRMS ETL pipeline, spatial mismatch handling and a deduplicated fire calendar. Ember Almanac describes a 25-year harmonized fire calendar. Treat this as confirmation that a plain fire map will not stand out; the differentiator needs to be scientific calibration, evidence, uncertainty, a clear user workflow, or a novel but defensible interaction.

Celestial X team page - Open link - Current 2026 team concept

Ember Almanac team page - Open link - Current 2026 team concept

Past Space Apps / adjacent precedent

Searches did not surface an earlier Space Apps challenge with this exact title. A related 2020 Space Apps project, "Mayas" for the Quiet Planet challenge, used VIIRS S-NPP active-fire information alongside settlement/ecological context. Use past fire applications primarily as UX inspiration; your 2026 requirement is specifically cross-sensor harmonization.

NASA Space Apps 2020 - Mayas / Quiet Planet - Open link - Historical Space Apps fire-data precedent

10. Kaggle, Hugging Face and Roboflow resources

10.1 Hugging Face - directly useful

Dataset	What is there	Use / caution	Link
LINKS hotspot-disambiguation dataset	2.23 GB; MODIS/VIIRS hotspots cross-referenced with EFFIS burned areas; ~500k stratified sample; features include FRP, temperatures, confidence, type, resolution, time, lat/lon; CC BY 4.0.	Useful optional classifier/feature-engineering reference for separating wildfire-like hotspots from other thermal anomalies. Do not treat Europe-trained labels as Bangladesh ground truth.	Open
Indonesian Wildfire Panel Dataset 2012-2025	0.25-degree monthly Sumatra/Kalimantan panel; VIIRS FIRMS + ERA5 + land/peat/concession features; Parquet; CC BY 4.0.	Very useful schema/modeling template for tropical fire data; geographic transfer should be treated cautiously.	Open
wildfire-dataset-pipeline	11 documented notebooks; 1 km daily grid combining open geospatial sources; parameterized Earth Engine extraction.	Good reusable ETL/grid pattern if you add prediction/context features later; not needed for the core harmonization MVP.	Open

10.2 Kaggle - convenience copies and notebooks

Kaggle is best used for quick exploration or notebook ideas; for the final challenge, cite and prefer NASA FIRMS as the authoritative source.

Vijayveer Singh Kaggle datasets page - Open link - Includes a dataset titled "NASA FIRMS Active Fire Data (MODIS/VIIRS)" (reported as 4 files / ~138 MB in indexed search results).

TabNet regressor notebook on NASA FIRMS data - Open link - ML/EDA inspiration; not a harmonization method by itself.

PCA notebook on NASA FIRMS data - Open link - Exploratory feature-analysis reference.

10.3 Roboflow Universe - optional computer vision

These image datasets are not substitutes for FIRMS hotspot records. Only use them if your team has finished the core calendar and wants an optional visual verification/classification module.

Dataset	Role	Link
Wildfire Satellite2	Satellite image classification/detection style dataset; community curated	Open
Wildfire Satellite	Community satellite wildfire image dataset	Open
Satellite Wildfire Detection	Small satellite wildfire detection dataset; useful only for CV prototype	Open

Priority rule Do not spend the hackathon training a satellite-image detector unless your calendar/harmonization is already complete. The official task is about reconciling MODIS and VIIRS hotspot records over time.

11. Research papers to read first

Paper	Title	Why it matters	Link
Giglio, Schroeder & Justice (2016)	The Collection 6 MODIS active fire detection algorithm and fire products	Explains MODIS C6 detection behavior, false alarms, probability of detection and product design.	Read
Schroeder et al. (2014)	The New VIIRS 375 m active fire detection data product	Foundational VIIRS 375 m algorithm paper; explains why VIIRS detects more/smaller fires and maps perimeters more coherently.	Read
Li et al. (2018)	Comparison of Fire Radiative Power Estimates From VIIRS and MODIS Observations	Critical harmonization evidence: FRP relationship varies with spatial scale, season/ecosystem and MODIS view angle.	Read
Xu et al. (2020)	Fire Detection and Fire Radiative Power in Forests and Low-Biomass Lands in Northeast Asia: MODIS versus VIIRS	Shows major detection differences, especially low-biomass/cropland, and demonstrates duplicate removal / near-simultaneous comparisons.	Read
Giglio et al. (2018)	The Collection 6 MODIS burned area mapping algorithm and product	Methodology behind MCD64A1, which is useful for secondary validation.	Read
Andela et al. (2019)	The Global Fire Atlas of individual fire size, duration, speed and direction	Event-level fire metrics and global data product built from MCD64A1; useful event benchmark.	Read
Artés et al. (2019)	A global wildfire dataset for the analysis of fire regimes and fire behaviour (GlobFire)	Global fire-event perimeters derived from MCD64A1; useful for cluster comparison.	Read

Paper	Title	Why it matters	Link
Urbanelli et al. (2023)	A Multimodal Supervised Machine Learning Approach for Satellite-based Wildfire Identification in Europe	Hotspot disambiguation using MODIS/VIIRS + burned-area labels; directly linked to the HF dataset.	Read
Artes et al./2024 active-fire dating validation	Active fire-based dating accuracy for Landsat burned area maps...	Shows agreement varies by biome, distance and fire size - a reminder not to assume one validation rule globally.	Read

What the literature implies for your method

- **Resolution bias is fundamental:** VIIRS 375 m has a greater response to relatively small fires than MODIS 1 km.
- **FRP helps but is not magically sensor-invariant:** published comparisons show FRP differences vary by scale, season, ecosystem and viewing geometry.
- **Use matched observations when possible:** near-simultaneous comparisons are stronger than arbitrary same-month point matching.
- **Land cover matters:** low-biomass/cropland and forests can show different omission behavior; stratification is defensible.
- **Aggregate before overfitting:** a stable 5 km/day or AOI/day comparison is usually more realistic for a four-day hackathon than pixel-perfect cross-sensor physics.
- **Show uncertainty:** sensor transfer is an estimation problem, not a perfect conversion.

12. Four-day execution plan

The submission screenshot provided by the team shows a 07 Oct 2026, 11:59 PM end. With this report dated 03 Oct, the plan below prioritizes a complete, defensible MVP before stretch features.

Date	Science/data	Backend/frontend	Demo/QA
Oct 3	Get MAP_KEY + Earthdata access; download Bangladesh standard FIRMS overlap; canonical schema; first raw comparison	Repo setup; AOI + map skeleton; data contracts	Finalize product name/story; assign roles; verify submission portal requirements
Oct 4	Build monthly/day aggregations; baseline calibration; holdout validation; exported Parquet/JSON	Calendar heatmap + Sensor Lab charts; API endpoints/cache	Capture validation screenshots; write methodology in README
Oct 5	Add land-cover stratification or event clustering only if baseline is stable	Connect AOI/map/calendar; provenance panel; responsive layout	Run end-to-end demo with fresh clone; source attributions
Oct 6	Freeze model unless a bug is found; one international contrast AOI	Polish, accessibility, no-data/error states; production deploy	Record full draft video; collect teammate feedback; repository cleanup
Oct 7	Only critical fixes; snapshot data/version	Freeze several hours before deadline; public source link test	Record final 3:45-3:55 video; test links in incognito; submit early; re-check portal timezone

Do not let perfect harmonization block a working demo A transparent robust baseline with measured error is stronger than an opaque advanced model that is not validated. Ship the baseline, then add sophistication behind a feature flag if it demonstrably helps.

13. Team roles and submission plan

Role	Owner focus	Definition of done
Science / harmonization lead	Sensor overlap, calibration model, validation, caveats	Reproducible notebook/script + metrics + method text

Role	Owner focus	Definition of done
Data / backend lead	FIRMS downloads, provenance, precompute, API/cache	One command builds compact data artifacts; API stays fast
Frontend / GIS lead	Map, AOI selection, calendar, sensor controls	Judges can select AOI and understand raw vs harmonized in <60s
UX / visualization + QA	Interaction flow, responsive mobile, legends, accessibility	No confusing raw/harmonized state; source/caveat labels present
Story / video / documentation	4-minute demo, README, impact, credits	Public links tested, claims match computed results, video under 240s

Suggested 240-second video

Time	Content
0:00-0:25	Problem: two sensor eras; 1 km MODIS and 375 m VIIRS raw counts are not directly comparable.
0:25-1:05	Select Bangladesh AOI. Show raw MODIS vs VIIRS overlap and the discontinuity.
1:05-1:50	Turn on harmonization. Explain overlap calibration and uncertainty in one sentence.
1:50-2:35	Open the 26-year calendar; click a high-activity month; show map and sensor provenance.
2:35-3:10	Show anomaly/percentile and one context/validation layer. Emphasize historical critical periods, not exact prediction.
3:10-3:35	Show Sensor Lab validation metric / holdout result and methodology/source panel.
3:35-4:00	Impact, global portability, open data/open source, and next step. End before the 240 s cap.

Submission hygiene checklist

- Public source-code link opens in an incognito browser without requesting permission.
- No API keys, MAP_KEY, Earthdata tokens, Hugging Face tokens or secrets are committed.
- README lists every dataset and license/attribution; NASA is not implied to endorse the app.
- Data build date, product versions and standard/NRT status are visible.
- Video is <=240 seconds and does not show features that are absent from the public build.
- Fallback/no-data states are tested before recording.
- Re-check the Bangladesh event portal for deadline/timezone changes immediately before submission.

14. Risks, caveats and scientific honesty

Risk	Why it matters	Mitigation
Hotspot != wildfire	FIRMS thermal anomalies may include agricultural burning, volcanoes, gas flares, hot smoke or other hot sources.	Use "hotspot" / "burning activity" by default; filter/flag type and persistent sources; only call an event wildfire when supported.
No detection != no fire	Cloud cover, overpass timing, sensor saturation/geometry and thresholds can hide activity.	Represent missing/low-coverage separately from zero; show data-quality flags.
Sensor resolution mismatch	VIIRS detects smaller hotspots than MODIS.	Calibrate in overlap period on common space/time units; never compare raw counts as a long trend.

Risk	Why it matters	Mitigation
FRP mismatch	FRP relationships vary by spatial scale, view angle and ecosystem.	Aggregate and validate; do not assume FRP is exactly interchangeable across sensors.
NRT vs standard	NRT is designed for timeliness and later replaced by standard quality.	Use standard for historical baseline; NRT only for recent operational panel and label it.
Type-field version issue	FIRMS reports a historical Collection-2 Vxx14IMGML Type-field bug in product versions 1/2, corrected and reprocessed to v3 in May 2025.	Use current processed versions; keep source version in provenance; avoid mixing old cached files.
S-NPP transition	FIRMS states S-NPP delivery ceases 01 Nov 2026.	Make VIIRS source list configurable; NOAA-20/21 should keep the product alive.
Validation dependence	MCD64A1 uses MODIS active-fire information, and several event datasets derive from MCD64A1.	Describe these as secondary consistency checks, not independent absolute truth.
Competition overlap	A same-challenge 2026 repository is already public.	Do independent work; use it only as a benchmark; cite any reused open-source components and comply with competition rules.

15. Clickable source directory

This directory consolidates the links scattered throughout the report. "Core" means directly relevant to the official challenge; "support" means useful for validation, context, benchmarking or implementation.

Official challenge and submission context

Official NASA Space Apps challenge - Open link

Bangladesh local portal (verify current submission details) - Open link

NASA FIRMS - core

FIRMS home - Open link

Active Fire downloads - Open link

Archive Download - Open link

Area API - Open link

MAP_KEY - Open link

Data Availability API - Open link

FIRMS API tutorial - Open link

Fire Data Academy - Open link

MODIS hotspot field definitions - Open link

VIIRS hotspot field definitions - Open link

FIRMS GIBS tutorial - Open link

FIRMS QGIS tutorial - Open link

FIRMS ArcGIS tutorial - Open link

NASA visualization and Earth Engine

NASA GIBS API docs - Open link

GIBS Access Basics - Open link

NASA Worldview - Open link

Earth Engine MODIS FIRMS NRT - Open link

Earth Engine S-NPP VIIRS NRT - Open link

Earth Engine NOAA-20 VIIRS NRT - Open link

Burned area / validation / context

MCD64A1 V6.1 - Open link
MCD64A1 DOI - Open link
VNP64A1 V2 - Open link
VNP64A1 DOI - Open link
MCD12Q1 land cover - Open link
MOD11A1 LST - Open link
VNP21A1D LST - Open link
NASA EONET v3 - Open link
GPM IMERG - Open link
NASA-USDA soil moisture - Open link
MERRA-2 - Open link
NASA POWER daily API - Open link

External public geospatial/fire data

GFED5 data - Open link
Global Fire Atlas DOI - Open link
Global Fire Atlas paper - Open link
GlobFire v2 - Open link
ESA WorldCover - Open link
geoBoundaries API - Open link
WorldPop data - Open link
ERA5-Land - Open link

Hugging Face / Kaggle / Roboflow

HF: hotspot-disambiguation dataset - Open link
HF: Indonesian Wildfire Panel - Open link
HF: wildfire dataset pipeline - Open link
Kaggle datasets by Vijayveer Singh - Open link
Kaggle TabNet FIRMS notebook - Open link
Kaggle PCA FIRMS notebook - Open link
Roboflow Wildfire Satellite2 - Open link
Roboflow Wildfire Satellite - Open link
Roboflow Satellite Wildfire Detection - Open link

Open-source projects

IGNIS - current 2026 same-challenge public submission - Open link
MODIS-VIIRS-fires (Meredith Franklin) - Open link
hotspot-disambiguation (LINKS) - Open link
wildfire dataset pipeline (JasmineW6214) - Open link
viirs-modis-fire-map - Open link
fireMap - Open link
FireSight - Open link
Call for Code fires-api-nodejs - Open link
fire-map (kjaymiller) - Open link

Research papers

The Collection 6 MODIS active fire detection algorithm and fire products - Open link

The New VIIRS 375 m active fire detection data product - Open link

Comparison of Fire Radiative Power Estimates From VIIRS and MODIS Observations - Open link

Fire Detection and Fire Radiative Power in Forests and Low-Biomass Lands in Northeast Asia: MODIS versus VIIRS - Open link

The Collection 6 MODIS burned area mapping algorithm and product - Open link

The Global Fire Atlas of individual fire size, duration, speed and direction - Open link

A global wildfire dataset for the analysis of fire regimes and fire behaviour (GlobFire) - Open link

A Multimodal Supervised Machine Learning Approach for Satellite-based Wildfire Identification in Europe - Open link

Active fire-based dating accuracy for Landsat burned area maps... - Open link

Final recommended scope

If you build only five things (1) a defensible cross-sensor calibration using the overlap period, (2) a 26-year burning calendar, (3) a raw-vs-harmonized Sensor Lab, (4) a map that drills into calendar cells, and (5) a provenance/uncertainty panel. Everything else is optional.

This keeps the project faithful to the official challenge, scientifically explainable, demonstrable in under four minutes, and realistic within the remaining hackathon window.